



A passion fruit counting method based on the lightweight YOLOv5s and improved DeepSORT

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Abstract

Accurate yield estimation of passion fruits is essential for planning acreage and harvest timing. However, due to the complexity of the natural environment and tracking instability, the existing yield estimation methods suffer from excessively large models that are difficult to deploy or repetitive counting of fruit. Therefore, an improved approach for efficient passion fruit yield estimation was proposed using the lightweight YOLOv5s and improved DeepSORT. First, the video is fed into the proposed lightweight YOLOv5s called YOLOv5s-little to obtain coordinates and confidence information about the fruits within each frame. Then, the information obtained from the detection model is input into improved DeepSORT for continuous frame tracking of passion fruit. Considering the frequent error IDs (ID switching), two improvements based on DeepSORT are proposed: delaying the creation of tracks and adding a second round of IoU matching. Finally, to overcome the problem of repetitive counting, a specific tracking counting method based on the track information and state is used for accurate passion fruit counting. Our method achieved a competitive result in tests. YOLOv5s-little detector achieved precision of 98.9%, 98.3% recall, 99.5% mAP, and only 0.9MB model size. The improved DeepSORT algorithm achieved higher order tracking accuracy (HOTA) of 79.6%, multi-object tracking accuracy (MOTA) of 92.58%, identification F1 (IDF1) of 95.02%, and ID switch (IDSW) of 11 respectively. Compared with DeepSORT, it improved by 4.66%, 1.8%, and 9.16% in HOTA, MOTA and IDF1, respectively, and IDSW improved the most with 85%. Compared with FairMOT and TransTrack, the HOTA of YOLOv5s-little + improved DeepSORT achieved improvements of 11.56% and 25.24%, respectively. The statistical average counting accuracy of our proposed counting method reaches 95.1%, which is a 7.09% improvement over the maximum ID value counting method. The counting results from test videos are highly correlated with the manual counting results ($R^2=0.96$), indicating that the counting method has high accuracy and effectiveness. These results show that YOLOv5s-little + improved DeepSORT can meet the practical needs of passion fruit yield estimation in real scenarios.

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Introduction

Passion fruit is rich in nutritional and medicinal value, which is mainly cultivated in tropical and subtropical regions. It is an important local crop in South America with high economic value (Bezerra et al., 2019; Thokchom & Mandal, 2017). However, manual yield estimation of passion fruit is a laborious and time-consuming task. This presents challenges for farmers in planning cultivation areas, determining optimal harvest times, and devising sales strategies. Fortunately, recent advancements in computer vision technology have provided effective support in addressing this issue (He et al., 2022a, 2022b; Vasconez et al., 2020). An automatic counting system for passion fruits can be developed, enabling farmers to plan their production and marketing strategies more accurately and maximizing their profitability.

With the development of deep learning methods, many researchers have implemented fruit detection and counting with the help of deep learning-based object detection (Assunção et al., 2022; Liu et al., 2017; Wang et al., 2022b). For example, Koirala et al. (2019) developed a new detection model called MangoYOLO based on the features of YOLOv2 (tiny) and YOLOv3. Tu et al. (2020) were able to robustly detect small fruits of passion fruit by feeding RGB-D images into an improved multiple scale faster region-based convolutional neural networks (MS-FRCNN) based model. Qi et al. (2022) used YOLOv5 as a target detection model to detect the main stem of litchi in litchi images and obtain the accurate location information of the main stem picking point. Liu et al. (2023) proposed a method based on binocular stereo vision and an improved YOLOv3 model to detect pineapples, improved by the optimization of the backbone network and the use of the binocular camera to get the left and right images. The above methods are used to detect fruits in capturing images for counting. However, the rapid development and widespread use of robots, mobile terminals, and smart devices in modern agriculture. Many detection models cannot be deployed on these devices due to limitations in calculating ability and storage capacity (He et al., 2022a, 2022b).

Therefore, model lightweight has become a trend in fruit detection. In particular, Zhou et al. (2020) used SSD with two lightweight backbones MobileNetV2 and InceptionV3 for kiwi detection in the field. Wang and He (2021) developed a small model size based on a channel-pruned YOLOv5s deep learning algorithm to accurately detect apple small fruit. Wang et al., (2022a, 2022b) proposed an improved YOLOv5 model to detect litchi, the improved backbone network uses ShuffleNetV2, introduces the CBAM module in the feature fusion stage, and adjusts the input size to 1280×1280 . Shen et al. (2023) obtained a lighter-weight YOLOv5s cluster detection model based on the channel pruning algorithm, and soft non-maximum suppression is introduced in the prediction stage to improve the detection performance of grape overlapping clusters. Inspired by the above studies, a lightweight network based on YOLOv5 was designed according to the characteristics of passion fruit.

The input from object detection for counting fruits is images. These counting methods can only detect a portion of the fruits in that image, and each image has overlapping parts. Therefore, these methods will inevitably double count, leading to inaccurate results. To address the issue, most researchers use the multi-object tracking (MOT) technique to perform fruit counting (Egi et al., 2022; He et al., 2022a, 2022b; Liu et al.,

2019). MOT locates targets in a video and assigns a unique ID to each object (Guo et al., 2022; Luo et al., 2021). For example, Gao et al. (2022) used a YOLOv4-tiny network integrated with the CSR-DCF algorithm to count apples, which solved the problem of losing targets during the tracking process. Tan et al. (2022) used an optical flow-based tracking method to estimate camera motions. The number of cotton seedlings was updated by comparing the positions of bounding boxes predicted by optical flow and detected by the YOLOv4 network in the same frame. Yang et al. (2022) used CenterNet to detect cotton seedlings, extract their identity embedding, and associate the data based on DeepSORT fused localization and identity information.

In addition, there have been studies combining segmentation and multi-objective tracking for fruit tracking. De Jong et al. (2022) used MOTS architectures called TrackR-CNN and PointTrack and demonstrated the applicability of these methods for the joint detection of apples. Ariza-Sentís et al. (2023) applied the PointTrack algorithm for the detection and tracking of grape bunches and compared two instance segmentation algorithms (YOLACT and spatial embedding) to find the most suitable method for detecting grape bunches.

However, due to the complexity of the natural environment, there are frequent error IDs during the tracking process, leading to double counting. Regarding this issue, many researchers have already conducted studies on it. For example, Zhang et al. (2022) proposed OrangeSort to establish a specific tracking region counting strategy and tracking algorithm based on motion displacement estimation, aiming to alleviate the double counting problem associated with shaded fruits. Ge et al. (2022) obtained tracking results and used OpenCV to create a virtual counting line to count tomatoes. Rong et al. (2023) designed a specific tracking region counting method to overcome the problem of tracked tomato cluster IDs. Since passion fruits are often partially obscured by leaves or other fruits as they grow, and some of them are small, this can cause them to be easily missed. If the fruit is missed for a long time, it may lose its original ID and be re-identified as a new ID. This will cause duplicate counting problems in subsequent counting tasks. However, Current solutions are complex and not well suited to the problem of tracking passion fruit. Therefore, we design a method to alleviate frequent error IDs.

In addition to the challenges of models that are too large to deploy and frequent error IDs, we found that the currently proposed counting methods suffer from significant errors or cumbersome processes. Presently, the widely adopted approach is to use the maximum ID value generated by DeepSORT as the predicted value. While easily obtainable, the maximum ID often deviates significantly from manual counts. To address this issue, some studies have introduced alternative methods. Parico and Ahamed (2021) compared the ROI method with the maximum ID way. The ROI method counts fruits by crossing a line horizontally. Tests have shown that this method brings predicted counts closer to manual counts. Although the count line method has a high accuracy rate, it requires additional steps to be performed after tracking.

To address these challenges mentioned above, we have proposed an improved passion fruit yield estimation method based on YOLOv5s and DeepSORT. The specific improvements we have made are as follows:

- (1) A lightweight network based on YOLOv5s is designed.
- (2) We have introduced a delaying the creation of tracks method to avoid frequent error IDs and added a second round of IoU matching to improve tracking accuracy.

- (3) We have proposed a convenient counting method based on track state and track information. We ensure that the estimated values are closer to the actual values and simplify the counting process by counting the fruits during the process of track creation and deletion.

Materials and methods

Dataset description

The videos used in the experiment were obtained in passion fruit plantations in Heyuan City, Guangdong Province, China, and Huadu District, Guangzhou City, China. The reason for choosing these two locations is that they represent typical characteristics of passion fruit cultivation. Both selected data collection sites are large passion fruit plantations using the widely used pergola cultivation technique. The video capture device was a cell phone (Huawei Mate30), and the horizontal shot was used. The distance between the capture device and the passion fruit was kept at 5–8 m in order to effectively capture multiple passion fruit plants and to ensure the passion fruit in the video is at the right size. In this acquisition, a total of 12 videos were captured, each video was about 15 s in length. The videos were stored in MP4 format with a frame rate of 25 fps and a resolution of 1280×720 . Figure 1 illustrates a partial dataset.

The dataset processing mainly includes video annotation, video cropping into images, and data augmentation. The following are the specific steps:

- (1) The videos of 15s with dense fruit content were chosen as the dataset. The length of the videos ensures that each video shows a portion of the characteristics of the passion fruit orchard and meets the criteria for fruit-densely (videos with 10 or more fruits). Then, each video was cut using FFmpeg software, and the captured videos were labeled



Fig. 1 Part of the dataset



Fig. 2 Annotated images of video sequence

using DrakLabel software. All the passion fruits in each frame image are annotated and each passion fruit is given a unique ID. Figure 2 shows an annotated image of the video.

- (2) After labeling the video, the videos were converted into image frames. A total of 1 287 image frames were obtained. These images were used for the training of YOLOv5 with DeepSORT. Additionally, a Python script was employed to transform the annotated files into txt files suitable for detection (COCO dataset) and reidentification (ReID) data suitable for DeepSORT. All image frames are randomly divided into training set, validation, and test set in the ratio of 7:1:2(900:130:257).
- (3) When performing data augmentation on the original image for the training set, we perform the following steps, including randomly adjusting brightness and contrast, adding Gaussian noise, and cropping the image. Finally, the original 900 images used for YOLOv5 training were expanded to 1800 images.

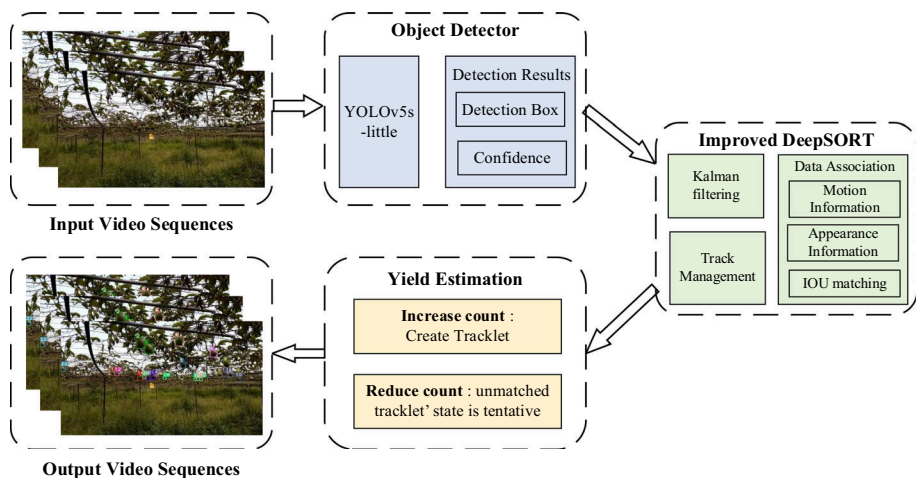


Fig. 3 Passion fruit yield estimation tracking flowchart

Method process and flow chart

Figure 3 illustrates the flowchart detailing the passion fruit yield estimation algorithm. The algorithm encompasses three main steps, each serving a crucial role in the overall process.

Object Detection using YOLOv5s-little: A lightweight detector named YOLOv5s-little is used to detect each passion fruit and obtain its detection box and confidence in each frame of the video.

Tracking with Improved DeepSORT: Information from the detector is imported into the improved DeepSORT algorithm to achieve accurate tracking of each passion fruit in the current frame and the next frame. To avoid error frequent IDs and improve tracking accuracy, two strategies in the improved DeepSORT are introduced: delaying the creation of track and adding the second round of IoU matching.

Passion Fruit Yield Estimation: The counting module completes passion fruit yield estimation according to track information and track state. The output includes the results of the MOT sequences and the passion fruit yield estimation.

YOLOv5s, YOLOv5s-2level and YOLOv5s-little detection network

As one of the most widely used one-stage object detection algorithms, YOLO has been widely applied in industry and agriculture due to its outstanding performance and fast detection speed (Bochkovskiy et al., 2020; Redmon & Farhadi, 2018). YOLOv5 was developed by Ultralytics and released in June 2020 as an upgrade to the YOLOv4 model. Compared to the structures of YOLOv3 and YOLOv4, YOLOv5 has been further optimized to enable fast and accurate object detection. In addition, YOLOv5's tractable network architecture allows for further optimization and enhancement, while also being conducive to use with other multi-target tracking algorithms. The YOLOv5 series includes four versions, YOLOv5s, YOLOv5m, YOLOv5x, and YOLOv5xl, which vary in network depth and width. YOLOv5s was chosen as the detector for improvement because it can balance the requirements of speed and accuracy.

Two improved YOLOv5 detection networks for passion fruit targets are proposed, the two-layer (YOLOv5s-2Level) and the lightweight network (YOLOv5s-little). YOLOv5s-little adopts the same structure as YOLOv5s-2Level. The network architecture of YOLOv5s and YOLOv5s-2Level is shown in Fig. 4. As the targets of passion fruit in video tracking are generally small, the YOLOv5s model has little impact on the accuracy of fruit detection in high-level feature maps for large targets, and it consumes more detection time. Therefore, YOLOv5s-2Level mainly removes the P5 layer in the YOLOv5s model. Since passion fruits are usually round and have little shape variation, based on YOLOv5s-2Level, YOLOv5s-little

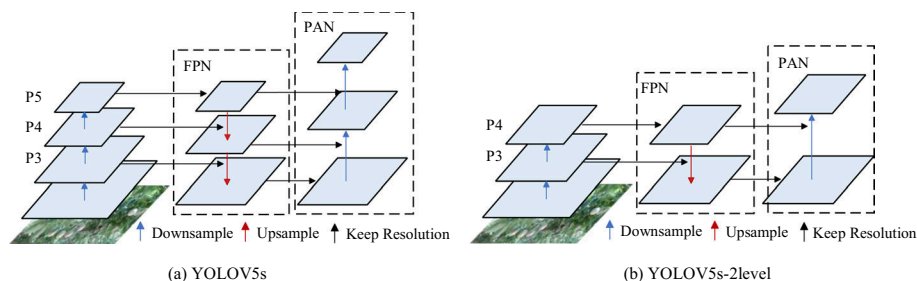


Fig. 4 Network structure of YOLOv5s and YOLOv5s-2level

further decreases model parameters, compresses network depth and width, and reduces the number of anchor boxes to shrink the model size, making it less than one-tenth the size of the YOLOv5s model while maintaining fast detection speed and good detection accuracy.

Improved deepSORT for passion fruit yield estimation model

The DeepSORT algorithm is an improved algorithm based on the SORT algorithm (Wojke et al., 2017). The SORT algorithm consists of three parts, Kalman filtering, the Hungarian matching algorithm, and track management (Bewley et al., 2016). The algorithm first uses Kalman filtering (KF) to predict the position of each detected target. Then a data association algorithm is used to associate the tracked targets with their detection results in consecutive frames. Finally, tracks on matches are updated, tracks that are no longer detected are deleted, and new tracks are created for unmatched objects. On top of the SORT, DeepSORT uses a convolutional neural network to extract the features of the target and employs appearance similarity for data association. This enhances the precision and robustness of multi-object tracking, particularly in complex scenarios.

Figure 5 shows the flowchart of the improved DeepSORT algorithm. The specific tracking process is as follows:

- (1) First, cascade matching is performed between the detection results and the confirmed state track predicted by the KF. The Hungarian matching algorithm is used to match the previous tracks with the current detection boxes by solving the cost matrix of motion and appearance information.
- (2) For unmatched detection boxes, unmatched tracks, and unconfirmed tracks, the algorithm uses first round of IoU matching to complete trajectory tracking using the Hungarian algorithm.
- (3) The second round of IoU matching is performed to match the unmatched tracks and detection boxes from the first round of IoU matching. If an unmatched detection satisfies the track creation condition, a track is created. If an unmatched track satisfies the deletion condition, the track is deleted. If an unmatched track does not satisfy the deletion condition, it continues to participate in the next round of matching.
- (4) After all matching in the current frame is completed, the KF update is performed on the matched tracks.

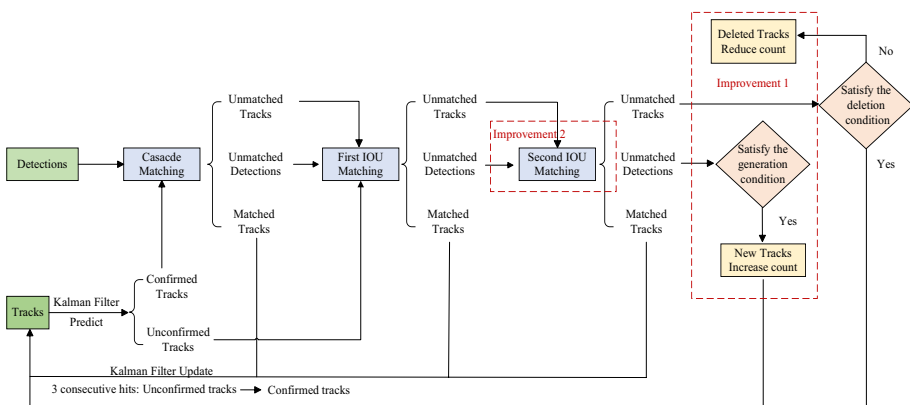


Fig. 5 Flow chart of the tracking process of the improved DeepSORT



Fig. 6 Illustration of passion fruit frequent IDs

Fig. 7 Area map allowing creation of tracks



Two improvements to DeepSORT (as shown in Fig. 5) are proposed to address the problems encountered in passion fruit tracking, namely delaying the creation of tracks and the second round of IoU matching. They are described below:

(1) Delaying the creation of tracks

Due to the mutual occlusion between fruits and the similarity in color between immature fruits and leaves, it is easy to cause false detection of fruits, leading to frequent IDs. Figure 6 displays the frequent IDs of passion fruit in real-life scenarios. In Fig. 6a, the highest ID shown is 47, whereas the manual annotation for this frame indicates 27. Similarly, in Fig. 6b, the highest ID is 21, but the manual annotation indicates 19. After many experiments, we find that frequent error IDs occur mainly in the central region and the border region. The reasons for this include two points. First, fruits in the central region lose their original ID due to feature significant changes (e.g., illumination, shape, occlusion, etc.) and are assigned a new ID. Second, leaves and stems in the border region are easily error detected as fruits, leading to error frequent IDs.

Therefore, we propose a method for the creation of tracks to limit the increase in fruit ID in the central and boundary regions. This method is shown as improvement 1 in Fig. 5. Specifically, in Fig. 7, the inner red area is the central area, and the outer red area is the boundary area. New tracks can only be created in the red area, not in the center area or the boundary area. During the experiment, we use the two values of coordinates of the bounding box (BB) and the resolution of the current frame to determine whether the bounding box is in the central region or the boundary region. After the third frame, if the unmatched detection is in the central area or the boundary area, no track is allowed to be created,

otherwise a track is created and given an ID. The boundary value is set according to the size of the fruit (set 25 in this paper). The detailed steps are described in Algorithm 1.

Algorithm 1 The creation of track

Input: [x, y, w, h]: bounding box information

frame_idx: current frame index

img_width: current frame width

img_height: current frame height

next_id: next ID number

total: the total number of passion fruits

Output: next_id; total

```

1: if frame_idx > 3 then
2:   if w < 25 and (0 < x < 30 or img_width - 30 < x < img_width) then
3:     return
4:   end if
5:   else if 70 < x < img_width - 70 - w and 70 < y < (img_height - 70 - h) then
6:     return
7:   end if
8: end if
9: create_track (x, y, w, h, next_id)
10: next_id ← next_id + 1
11: total ← total + 1
12: return next_id, total

```

(2) The second round of IoU matching

Due to delaying the creation of tracks, new tracks cannot be created in either the central region or the boundary region. If a track in these regions fails to match new detection boxes multiple times, the track is then deleted. In addition, no new tracks could be created for fruits in these regions, leading to missed detections. Therefore, we add the second round of IoU matching to achieve better results in associating unmatched detections with tracks that were not matched in the first two rounds of matching.

Yield estimation algorithm

The maximum ID value generated based on DeepSORT can be used to predict the number of passion fruits, and the maximum ID value counting method (MAX-IDVC) is commonly used. However, the predicted count from MAX-IDVC exceeds the actual count. The single line counting method, which counts fruits by tracking them as they cross a line horizontally, can solve the repetition counting problem. But it needs to be counted after each frame of tracking, which is a rather cumbersome process. Our proposed method estimates the yield of fruits based on the track information and state during the track management process. This method not only resolves duplicate counting problems but also simplifies operations. This yield estimation algorithm is shown as improvement 1 in Fig. 6.

The passion fruit yield count of each video is set as Total. The yield count is estimated in three cases based on track information and track state, specifically as shown in Eq. 1. Total is added by 1 if there is a creation of the track. The total is subtracted by 1 for the unmatched track state is tentative. The total remains unchanged if the number of recent KF update (time-since-update) parameters of the track is greater than 30.

$$\text{Total} = \begin{cases} \text{Total} + 1 & \text{if a creation of track} \\ \text{Total} - 1 & \text{if unmatched track state is tentative} \\ \text{Total} & \text{if track's time-since-update} > 30 \end{cases} \quad (1)$$

Experiment and result analysis

Implementation details

The experiments for estimating the passion fruit yield in this study are divided into three parts:

- YOLOv5s, YOLOv5s-2level, and YOLOv5s-little models were used to detect passion fruit.
- Based on the YOLOv5s-little model, passion fruit was tracked using DeepSORT and the improved DeepSORT.
- Based on YOLOv5s-little and the improved DeepSORT, MAX-IDVC, and our counting method were used to count passion fruits.

For the detection experiment, 4000 images with data augmentation were used for model training. The original resolution of the images was 1280×720 pixels, but it was compressed to 640×640 pixels after input preprocessing. In the training process, a pre-trained

model was not used, and the optimizer was set to SGD. The learning rate, batch size, and epoch were set to 0.01, 32, and 200, respectively.

Twelve annotated videos were used, with three videos used for tracking testing and the remaining nine videos used for ReID training. The target dataset for ReID training contained 198 passion fruits. The number of images per passion fruit was increased to around 250 through data augmentation. The ratio of the training set to the test set was set at 4:1.

For the tracking experiment, three videos with 28, 42, and 19 passion fruits, respectively, were used, covering both sparse and dense scenes. The max_IoU_distance threshold was set to 0.7, while other parameters were kept at their default values.

Evaluation metrics

In this section, we have chosen metrics because of their widespread use and comprehensive nature, which allows for effective comparisons with other studies and ensures the credibility and comparability of our research. The evaluation metrics used in each of the three experiments are described below.

The object detection section was evaluated using precision (P), recall (R), mean average precision (mAP), run time, and model size for YOLOv5s, YOLOv5s-2Level, and YOLOv5s-little (Padilla et al., 2020). Among them, P, R and, mAP reflect the overall performance of the detector, and the higher the value, the better the performance. The running time and model size reflect the lightness of the detector, and the lower the value, the higher the lightness.

The multi-object tracking part uses higher order tracking accuracy (HOTA), multiple object tracking accuracy (MOTA), identification F1 (IDF1), ID switch (IDSW) for DeepSORT and the improved DeepSORT is evaluated comprehensively (Luiten et al., 2021). Among them, HOTA, MOTA, IDF1, and IDSW are important to reflect the performance of the tracking algorithm, the higher the value of HOTA, MOTA, and IDF1, the better the performance, and the smaller the value of IDSW, the better the performance.

The counting part uses counting accuracy and Statistical average counting accuracy (Rong et al., 2023) to evaluate the counting method. The higher the value of these two metrics, the closer the predicted value of the counting method is to the true value.

The detection results of YOLOv5s, YOLOv5s-2Level, and YOLOv5s-little

Table 1 displayed the evaluation results of the YOLOv5s, YOLOv5s-2Level, and YOLOv5s-little detectors. The main difference in precision between the three models is reflected in the values of mAP@0.5:0.95. And the mAP@0.5:0.95 values for YOLOv5s-2Level and YOLOv5s-little were 0.856 and 0.782, respectively, representing a decrease of 1.4% and 8.8% compared to the original YOLOv5s model. Furthermore, the detection time

Table 1 Detection results of YOLOv5s, YOLOv5s-2Level, and YOLOv5s-little

	P/%	R/%	mAP@0.5/%	mAP@0.5:0.95/%	Time/s	Size/MB
YOLOv5s	99.1	98.8	99.8	87	0.0127	14.4
YOLOv5s-2Level	98.7	99.6	99.8	85.6	0.0097	3.4
YOLOv5s-little	98.9	98.3	99.5	78.2	0.0079	0.9

per image for YOLOv5s-2Level and YOLOv5s-little models were 0.0097 s and 0.0079 s, respectively, resulting in a speed improvement of 30% and 60%. With a size of less than 1MB, the YOLOv5s-little model was more suitable for online, real-time applications and portable devices than the YOLOv5s and YOLOv5s-2Level. Considering detection speed, precision, and model size, the YOLOv5s-little model was chosen as detector for the later tracking experiments.

Figure 8 showed the detection results for YOLOv5s and YOLOv5s-little under sparse (in the left column of Fig. 8) and dense scene (in the right column of Fig. 8). In the former, nine fruits were present in the original image, and both YOLOv5s and YOLOv5s-little successfully detected them. In the latter, the original image contained twenty fruits, and both models also detected all of them with equal accuracy. Taken together, these results demonstrated the robust detection capability of both YOLOv5s and YOLOv5s-little across diverse scene densities.



(a) Original images in sparsely-fruited scenes.
Ground labeled result : 9



(b) Original images in densely-fruited scenes.
Ground labeled result : 20



(c) Detection map of YOLOv5s in sparsely-fruited scenes.
Detection result : 9



(d) Detection map of YOLOv5s in densely-fruited scenes.
Detection result : 20



(e) Detection map of YOLOv5s-little in sparsely-fruited scenes.
Detection result : 9



(f) Detection map of YOLOv5s-little in densely-fruited scenes.
Detection result : 20

Fig. 8 The detection results of YOLOv5s and YOLOv5s-little in different scenes

The tracking results of improved and original DeepSORT Based on YOLOv5s-little

The comparison results between the improved and original DeepSORT tracking algorithms based on YOLOv5s-little were presented in Table 2. The average or total of each evaluation metric in all test videos is labeled in bold. Five metrics mentioned in Sect. 3.2 were used to evaluate the tracking performance. The values of the HOTA, MOTA, IDF1, and IDSW indicators were 74.94%, 90.78%, 85.86%, and 76, respectively using the original DeepSORT, while they were 79.6%, 92.58%, 95.02%, and 11, respectively using the improved DeepSORT. The improved DeepSORT showed excellent performance in the first four indicators, with improvements of 4.66%, 1.8%, 9.16%, and 85% in HOTA, MOTA, IDF1, and IDSW, respectively, compared to the original algorithm. The improvement in IDSW was most significant, effectively reducing the problem of frequent IDs. This was very helpful for subsequent fruit-counting tasks and would lead to more accurate fruit-counting results, thus improving the efficiency and accuracy of crop management. Despite using IOU matching one more time than the original, the tracking speed of the improved DeepSORT did not decrease significantly. In conclusion, the improved DeepSORT showed significant improvements in both tracking stability and accuracy.

Figures 9 and 10 showed the tracking results of test videos 02 and 03 using the original and improved DeepSORT. Both videos had 100 frames. Test video 02 contained 19 fruits which were sparsely distributed, and video sequences were captured from right to left. Test video 3 contained 42 fruits which are densely distributed, while video 03 sequences were captured from right to left.

In Fig. 9a and c, the highest ID value by the original DeepSORT was 14, while the maximum ID value by the improved DeepSORT was 10. The original DeepSORT showed discontinuity in ID in the 50th frame, while the ID value by the improved DeepSORT remained continuous. In Fig. 9b and d, the maximum ID value by the improved DeepSORT was 19, while the maximum ID value by the original DeepSORT was 26. The maximum ID value by the original DeepSORT differed from the actual number of fruits by 7 units, while the improved DeepSORT was closer to the actual situation.

In Fig. 10a and c, the maximum ID value of the original DeepSORT reached 65, which exceeded the ground truth count significantly. However, the maximum ID value of the improved DeepSORT was 40, which was close to the ground truth count (equal to 42). In Fig. 10b and d, the maximum ID value of the original DeepSORT was 89, which was more than twice the ground truth count, while the maximum ID of the improved

Table 2 Tracking results of improved and original DeepSORT Based on YOLOv5s-little

Method	Test video	HOTA/%↑	MOTA/%↑	IDF1/%↑	IDSW↓	FPS ($f \cdot s^{-1}$)↑
Original DeepSORT	01	83.42	96.24	97.06	4	35
	02	66.53	87.58	75.05	71	52
	03	79.72	88.52	94.08	1	27
	All	74.94	90.78	85.86	76	38
Improved DeepSORT	01	84.25	96.73	98.27	2	33
	02	75.57	90.29	92.58	9	52
	03	80.69	90.46	95.25	0	28
	All	79.6	92.58	95.02	11	38

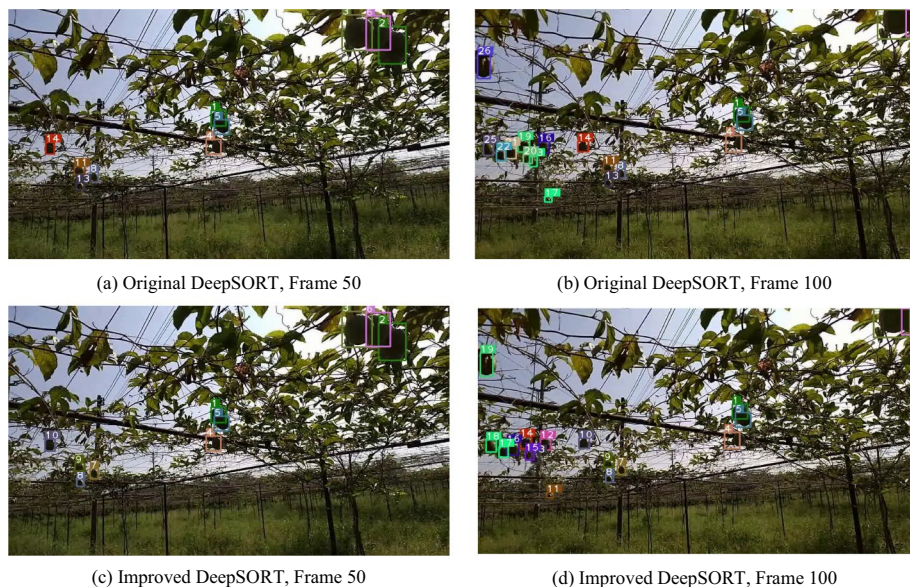


Fig. 9 The tracking results of original and improved DeepSORT of test 02

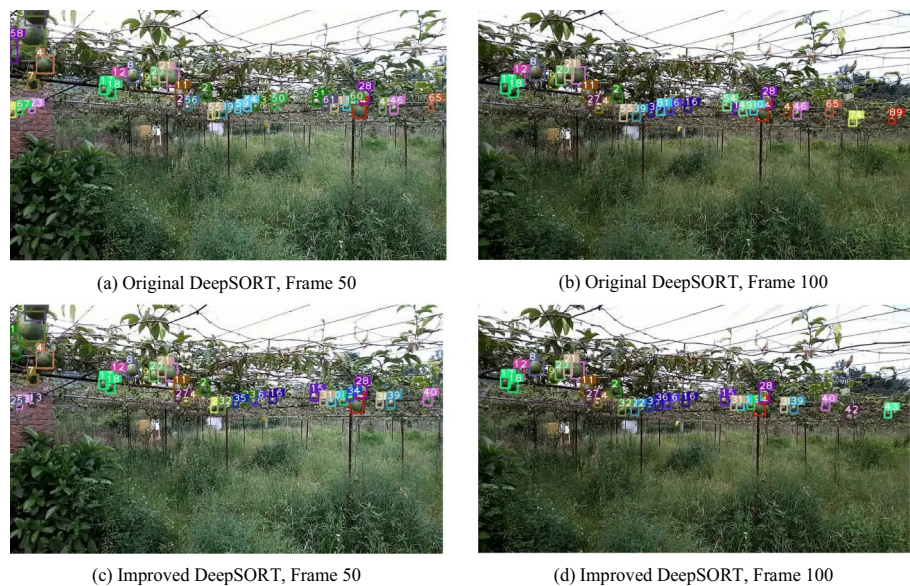


Fig. 10 The tracking results of original and improved DeepSORT of test 03

DeepSORT was 43, which was also close to the ground truth count. In conclusion, the improved DeepSORT can effectively solve the problem of frequent IDs in fruit sparse and dense scenarios.

Evaluation of counting method

The comparison experiments between the MAX-IDVC approach and our counting method were shown in Fig. 11. The x-axis in Fig. 11 represented the video number, and the y-axis represented the fruit counts. The blue bars represented the ground truth of the fruit count value, while the red bars represented the fruit count values between the two methods including the MAX-IDVC approach and our counting method. The counting accuracy percentage values were shown above each bar.

According to the results of Fig. 11, our counting method achieved higher counting accuracy than that of the MAX-IDVC method for most videos. In Fig. 11a, the lowest counting accuracy achieved by the MAX-IDVC method is 52.4%, while the highest counting accuracy was 94.7%. In Fig. 11b, our counting method achieved the lowest counting accuracy of 76.2% and the highest counting accuracy of 100%. Statistical average counting accuracy using the MAX-IDVC method was 86.19%, while statistical average counting accuracy using our counting method was 93.28%. Therefore, our counting method can achieve better result than that of the MAX-IDVC method.

The regression curves between the predicted values and the ground truth count between the maximum ID value counting and the proposed methods were shown in Fig. 12. Figure 12a displayed the regression curve between the maximum ID value and the ground truth count, while Fig. 12b showed the regression curve between the proposed method count value and the ground truth count. The R^2 value of the regression curve between the maximum ID value and the ground truth count was 0.9419, while the R^2 value of the proposed method was 0.9633, which was higher than that of the maximum ID value regression curve. In addition, the slope of the regression curve of the proposed method was closer to 1 than that of the maximum ID value. In conclusion, the proposed method was more accurate in predicting the passion fruit yield.

Comparison with other MOT method

FairMOT and TransTrack were chosen for comparison with lightweight YOLOv5s-little and improved DeepSORT. FairMOT (Zhang et al., 2021) consists of two homogeneous branches—one for predicting pixel-level object scores and the other for re-ID features. This

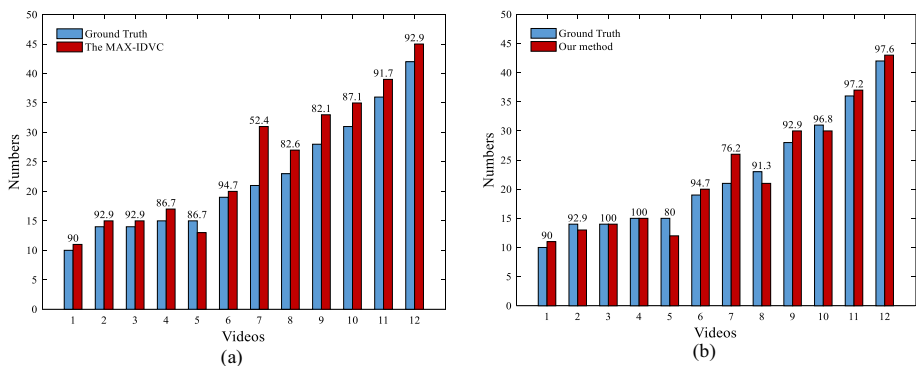


Fig. 11 The counting results of the ground truth and the predicted value. **a** The MAX-IDVC method. **b** Our counting method

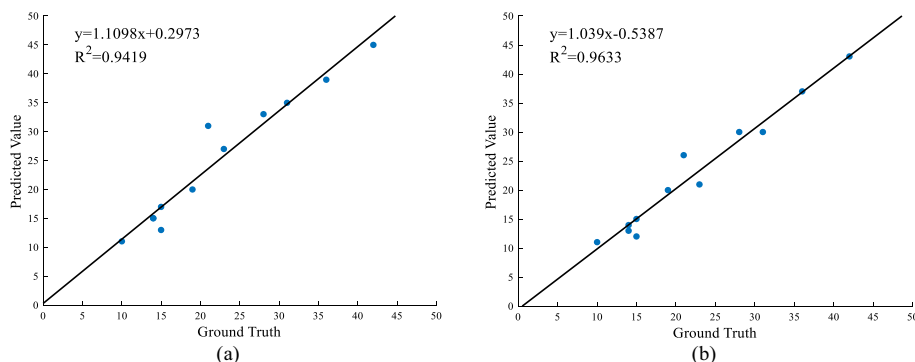


Fig. 12 The regression curve between the ground truth and the predicted value. **a** The MAX-IDVC method. **b** Our counting method

algorithm addresses the fairness issue between these two tasks and achieves high levels of detection and tracking accuracy. On the other hand, TransTrack (Sun et al., 2020) establishes a new joint detection and tracking paradigm. TransTrack fully leverages the advantages of transformer architecture to achieve object detection and object association in a single shot.

Table 3 shows the comparison of experimental results using our method with FairMOT and TransTrack tracking methods. The best value for each metric is shown in bold. Our method performs the best on four metrics of MOT. Specifically, the HOTA, MOTA, IDF1, and IDSW values of our approach were 79.6%, 92.58%, 95.02%, and 11, respectively. Compared to the FairMOT method, our method improved by 11.56% for HOTA, 7.91% for MOTA, and 19.29% for IDF1, while reducing IDSW by 269. Compared to TransTrack, our method achieved significant improvements of 25.24% for HOTA, 37.22% for MOTA, and 23.67% for IDF1. Additionally, the IDSW values of TransTrack were 351 higher than that of our approach. These comparative results further validated the effectiveness of our method in reducing frequent errors in fruit IDs and improving the accuracy of yield estimation algorithms.

Discussion

While existing fruit counting methods focus on clustered fruits (e.g., tomatoes, etc.), our study focuses on addressing the challenge of counting passion fruit individually. The characteristics of passion fruit make it similar in color to leaves and occlusion by leaves. To

Table 3 The experimental results of comparison with other tracking methods

Method	HOTA/%↑	MOTA/%↑	IDF1/%↑	IDSW↓
YOLOv5s-little + Improved DeepSORT	79.60	92.58	95.02	11
FairMOT	68.04	84.67	75.73	280
TransTrack	54.36	55.36	71.35	351

address these challenges, three key improvements for passion fruit counting are proposed: model lightweighting, delaying the creation of track and adding a second round of IoU matching to enhance tracking stability, and a counting method based on track information and track state. Our study shows that with these improvements, our proposed method makes significant progress in terms of model lightweight, tracking stability, and counting accuracy.

First, the YOLOv5s detection model is replaced by a lightweight detector named YOLOv5s-little. In the test set, YOLOv5s-little can effectively detect passion fruit in dense or sparse scenes, and the size of the model is only 0.9 MB. Compared to the lightweight networks such as MobileNet or ShuffleNet (Wang et al.,) based on YOLOv5s, the model of the proposed YOLOv5s-little is smaller in size. While the YOLOv5s-little performs well in that scene, it may not perform well when faced with different lighting or angle changes in different scenes. This may pose a challenge to its application in a wide range of scenes.

Second, we delay the creation of tracks and add the second round of IoU matching during passion fruit tracking, effectively reducing the frequent error IDs. Table 4 shows the results of the ablation study for the tracking improvement method. The best value for each metric is shown in bold. The original tracker achieved the HOTA of 74.94%, MOTA of 90.78%, IDF1 of 85.86%, and IDSW of 76. The value of HOTA and MOTA using the second round of IoU matching increases by 2.89% and 1.68% compared with the original tracker, which indicates that adopting it can get better tracking performance. Delaying the creation of tracks and the second round of IoU matching can further decrease the IDSW to 11. At the same time, the values of HOTA, MOTA, and IDF1 can also be enhanced by 4.66%, 1.8%, and 9.16% compared with the original one (Table 4 first row). Therefore, by reducing IDs and improving tracking performance, the improved DeepSORT can consistently track passion fruit in videos. These enhancements significantly contribute to the accuracy improvement in subsequent counting results. However, when the restricted area is set unreasonably, it limits the generation of new tracks within that area, which might lead to missed detections. Therefore, considering the characteristics of different scenarios, it's necessary to adjust the restricted area based on the characteristics of different scenes.

Finally, when using DeepSORT to track each passion fruit, calculating the total number of passion fruits directly based on the maximum ID value leads to inaccurate counting. To solve this problem, a counting method based on track information and track state is proposed to overcome the double counting problem, and the number of passion fruits using our method is close to manual counting.

The different classes of MOT algorithms currently available all have their own characteristics. For example, FairMOT has higher computational efficiency and is able to perform target tracking in real time, but its performance may be poor in complex scenarios. TransTrack is capable of handling large-scale target sets based on Transformer, it may face computational complexity and speed challenges and is not applicable in real

Table 4 Ablation study of tracking improvement

Delaying the creation of tracks	The second round of IoU matching	HOTA↑	MOTA↑	IDF1↑	IDSW↓
√	√	74.94	90.78	85.86	76
		74.87	86.52	87.94	29
√	√	77.83	92.46	91.30	33
		79.60	92.58	95.02	11

production environments. Compared to these two tracking algorithms, the proposed improved DeepSORT offers an improvement in tracking accuracy while satisfying better real-time performance and can be adapted for use in practical applications.

When estimating the passion fruit yield, there are two key challenges in video data collection. Firstly, rapid movement of the camera can result in blurred images, making it difficult to capture and detect the fruits. Hence, stabilizing the camera is necessary to ensure accurate fruit detection. Secondly, shooting toward sunlight can result in excessively dark footage, affecting image quality. To better track the fruits, video capture should avoid shooting toward sunlight.

In passion fruit yield estimation, it is difficult to fully consider agricultural expertise and specific environmental factors using only machine learning models. In subsequent studies, incorporating human expertise into machine learning processes, such as integrating the experience of agricultural experts, can significantly improve the accuracy and interpretability of prediction results (Holzinger et al., 2022). This approach not only provides yield-related data, but also delves into the factors that influence plant growth stages, including soil conditions, climate change, and plant pests and diseases. Such insights help to identify the true causality behind yield fluctuations, providing farmers with more targeted support and decision-making advice.

Conclusion

This study presents a method for estimating passion fruit yield by combining the lightweight YOLOv5s-little detector and an improved DeepSORT algorithm. By adopting a lightweight detection network and multi-object tracking techniques, our method achieves accurate passion fruit counting in natural environments. The YOLOv5s-little detector demonstrates excellent performance with 98.9% precision, 98.3% recall, and 99.5% mAP@0.5, while the model size is only 0.9MB. The improved DeepSORT algorithm achieves HOTA of 79.6%, MOTA of 92.58%, IDF1 of 95.02%, and IDSW of 11 respectively. A specific counting method based on track information and track state is proposed to overcome the issue of double counting, achieving a statistical average counting accuracy of 95.1%. And the counting results from test videos are highly correlated with the manual counting results ($R^2=0.96$).

The passion fruit yield estimation method proposed in this study will have a profound impact on the agricultural industry. Issues of model oversizing and double counting in existing methods have been successfully addressed, providing a more stable and real-time means of yield estimation for agricultural production. In the future, further optimization of the model's performance will be undertaken by considering additional factors in real-world planting scenarios, such as changes in light and fruit ripeness. Meanwhile, the integration of human expertise into the machine learning process is planned to enhance the accuracy and interpretability of prediction results.

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